Assembly guide

KEEP THESE INSTRUCTIONS FOR FUTURE REFERENCE

You printed a rocket. Now let's fly it!

This guide is for rockets made with **Rocket Builder** (rocket.quiverdesignworks.net). It shows what to do after the parts come off the printer: put them together, tie on the parachute, add a motor and launch.

Read each step all the way through before you start it. Work with a grown-up. Steps marked ADULT STEP are for the grown-up only.

The drawings show two sizes of rocket: an 18 mm (M18) rocket with one body piece, and a taller 24 mm (M24) one with two. Yours may have more or fewer body pieces. Drawings are not to scale.

WHAT YOU NEED

NOT IN THE BOX: MOTOR, IGNITER, LAUNCH PAD AND CONTROLLER



- Fine sandpaper — 220–400 grit
- Hobby knife — adults only
- Super glue (CA) — body joints only
- Ruler — in mm
- Pencil — to mark the balance point
- Scissors — cord, wadding
- Masking tape — fit fixes
- Stiff wire — or a floss threader
- Talc powder — optional, for the chute

WHAT HAPPENS IN A FLIGHT

1. The igniter lights the motor. The rocket slides up the launch rod.
2. The motor burns out. The rocket coasts up to the top of its flight.
3. The motor's **ejection charge** puffs gas up the body. It pushes the wadding, the parachute and the nose out of the top.
4. The nose stays tied to the body by the **shock cord**. The parachute opens. Everything floats down together.

HOW THE PARTS HOLD TOGETHER

- **Fins** are printed on the aft section. No fin gluing!
- **Body pieces** slide together on short shoulders. You glue these joints so they stay shut.
- **The nose** slides in too, but is **never glued**. It must pop off.
- **The motor** pushes against a ring at the top of its tube. The **motor cap** screws onto the tail and holds it in.

SAFETY FIRST

Model rocket motors and igniters are handled by an adult. Fly only in a big open field, with a launch pad and an electric launch controller, and follow the National Association of Rocketry (NAR) Model Rocket Safety Code (summary on page 7).

STEPS AT A GLANCE

1–4 Build p. 3
Clean up, test-fit, glue the body, tie the cord to the strap.

5–8 Recovery p. 4–5
Cord to the nose, parachute on, fold, pack.

9–12 Motor and pad p. 5–6
Adult steps: motor, cap, igniter, launch.

MY ROCKET

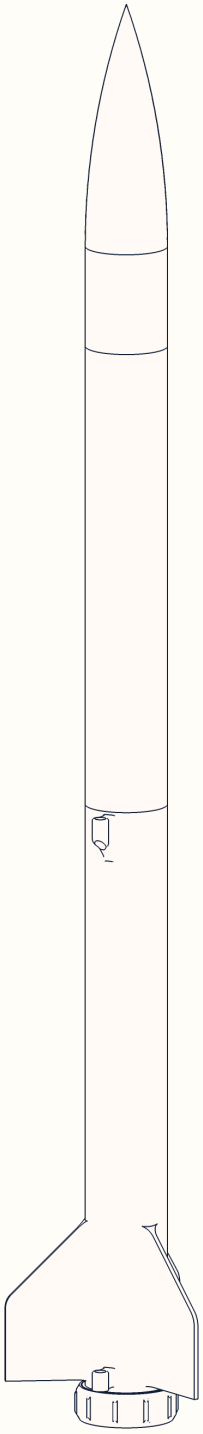
FILL THIS IN

NAME _____ BEST MOTOR _____

BUILDER _____ PARACHUTE _____

MOTOR SIZE _____ BALANCE POINT _____

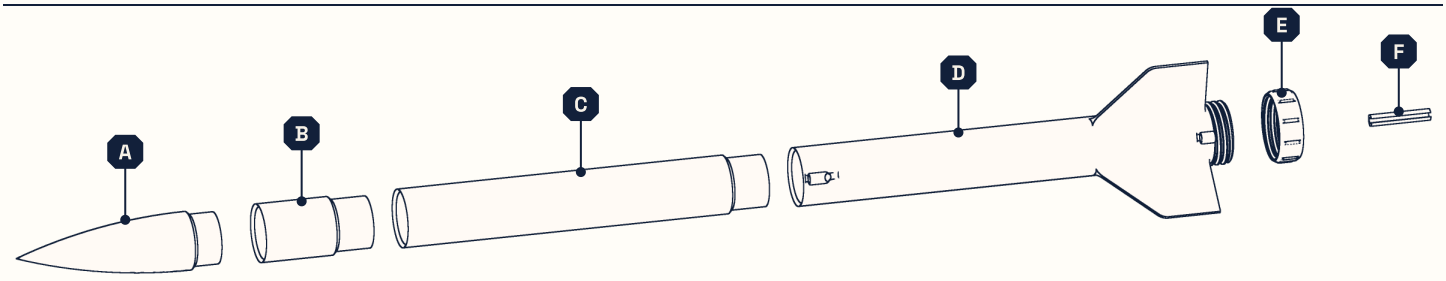
Copy these from your rocket's report in Rocket Builder or from *build_notes.txt* in your download. The balance point is the CG, in mm from the nose tip.



Parts layout

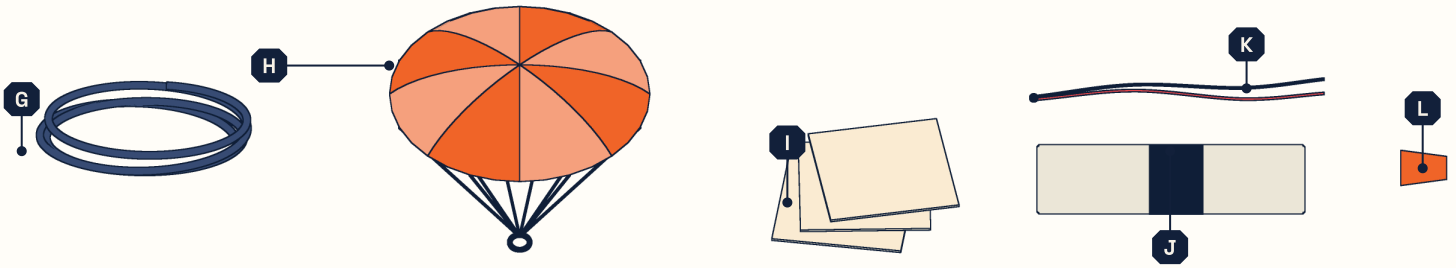
Lay out your parts and check them off. Each part has a letter. The steps use the same letters.

PRINTED PARTS — A–F



	PART	WHAT IT DOES	✓
A	**Nose cone**	Points the way. A short shoulder slides into the body. Inside the bottom is a little **tie bar** for the shock cord.	☐
B	**Anchor section**	The short body piece under the nose. A **strap** (a bar across the inside) holds the shock cord.	☐
C	**Body section** (C1, C2… or none)	Tall rockets are split into pieces that fit your printer. Short rockets have no C: B fits straight onto D.	☐
D	**Aft section**	The tail, with the fins, the launch lug(s) and the motor tube. The thread at the bottom is for the cap.	☐
E	**Motor cap**	Screws onto the tail and holds the motor in. The hole lets the motor's flame out.	☐
F	**Rod standoff**	A clip for the launch rod. It stays on the pad and holds the rocket up off the blast plate.	☐

YOU ADD — G–L



G	**Shock cord**	Thin, strong cord, about 2–3 times as long as the rocket. It must fit the 2.5 mm slot behind the strap: Kevlar or braided cord up to about 2 mm thick.
H	**Parachute** (or streamer)	Use the size your Rocket Builder report suggests (“Which parachute?”). Very light rockets can use a streamer.
I	**Recovery wadding**	Flame-resistant tissue. It keeps the hot ejection gas off the parachute. Use new wadding every flight.
J	**Motor** ADULT	A certified model rocket motor of your rocket's size (M13, M18, M24 or M29) and length. Use the motor your report recommends.
K L	**Igniter and plug** ADULT	Come with the motors. The plug holds the igniter in the nozzle.

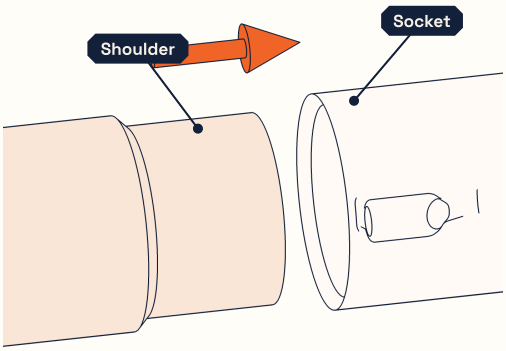
TIP

Not sure which motor or parachute? Open your rocket in Rocket Builder. The report shows the best motor, how high it goes, the parachute size and the launch field it needs.

1 Clean up the prints

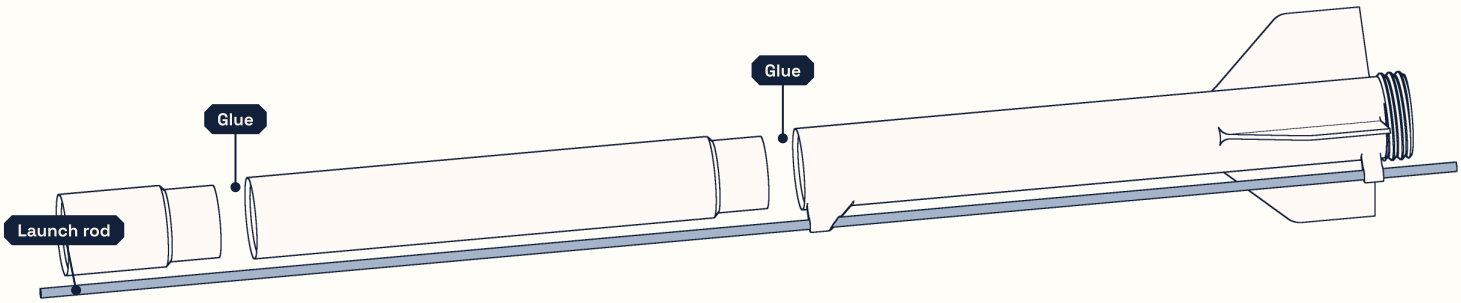
- Pull off any brim, strings and blobs.
- The edge that sat on the print bed can bulge out a little. Sand that lip off the **shoulders** (the narrow ends) and the start of the **tail thread**.
- Look for cracks, gaps between layers or thin spots, especially where the fins meet the tube. Don't fly a cracked part. Print it again.

2 Test-fit every joint (no glue yet)



Push each shoulder into the next part with a little twist. It should go all the way in, snug, with no gap at the seam. Too tight? Sand the shoulder lightly and try again.

3 Glue the body joints, with the launch lugs in line

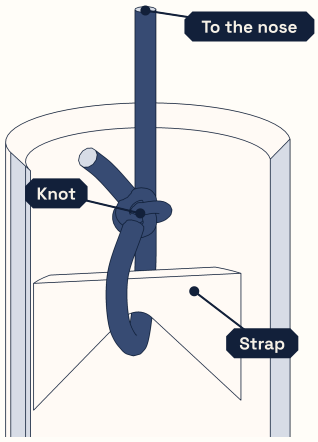


- Put a **thin ring of super glue** round the shoulder of C, near its step, and push it into D. Twist it a little so the glue spreads.
- If C or B has a launch lug, slide a launch rod through all the lugs while the glue sets, so they line up.
- Glue B onto C (or onto D if you have no C) the same way.

DO NOT GLUE

the nose (A) or the motor cap (E). The nose must pop off in flight, and the cap comes off after every flight. Keep glue out of the slot behind the strap inside B.

4 Tie the shock cord to the strap



1. Look into the top of B. The strap is a flat bar across one side, with a narrow slot behind it.
2. Push one end of the cord **down the slot behind the bar**. A bent wire helps. Pull the end out under the bar.
3. Bring it up **in front of the bar**. Tie it round the long part of the cord with two overhand knots, close to the bar.
4. Tug hard. The knot must not slip.

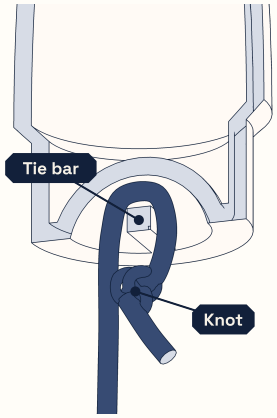
TIP

This is easier before B is glued on: you can reach in from both ends. The cut-away drawing has the front of B taken away so you can see the strap.

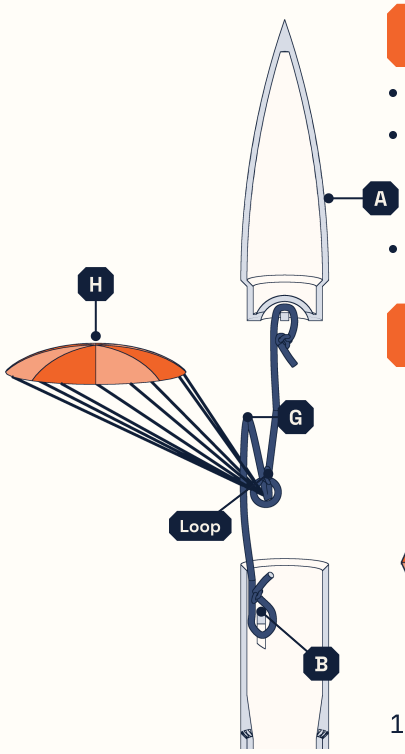
Rig the recovery system

The recovery system is the cord, the parachute and the wadding. It brings your rocket down gently, all in one piece.

5 Tie the other end of the cord to the nose



1. Turn the nose (A) over. A small **tie bar** crosses the little dome in the bottom of its shoulder.
2. Thread the cord's free end under the bar, round it and back out. A bent wire helps here too.
3. Tie it round the cord with two overhand knots.
4. Tug hard on the nose and the body. Nothing should slip.

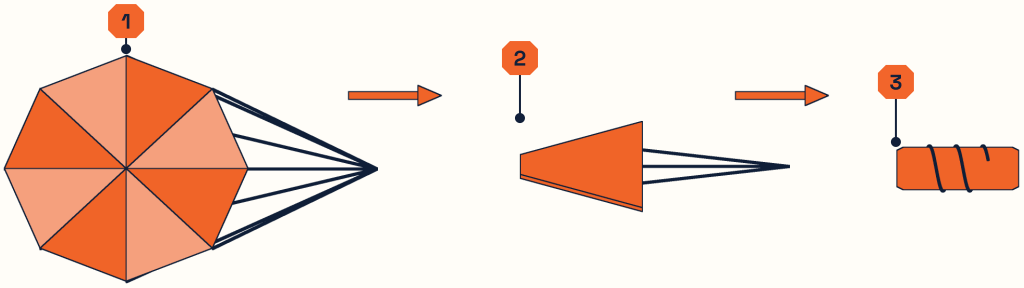


How it hangs together in the air. Not to scale.

6 Tie the parachute to the cord

- About a third of the way from the nose, tie a small **loop knot** in the cord.
- Gather the parachute's lines. Push their ends through the loop. Then pass the whole parachute through the lines' own loop and pull snug. (Or clip on a small snap swivel.)
- Now the nose, the parachute and the body all stay together.

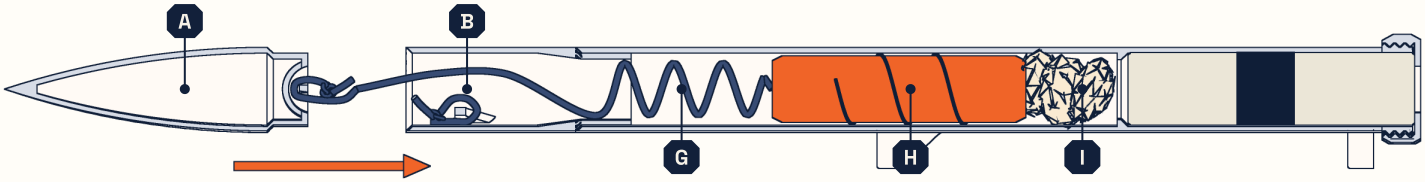
7 Fold and roll the parachute



1. Lay it flat. Pull the lines straight and together.
2. Fold it in half, then in half again, into a thin wedge. Lay the lines along it.
3. Roll it up loosely from the top and wrap the lines round the roll.

> **TIP**
> Do steps 7 and 8 before every flight. A pinch of talc keeps a plastic parachute from sticking on cold days. Use the size your Rocket Builder report suggests.

8 Pack it: wadding, parachute, cord, nose



1. Loosely crumple 3–4 squares of wadding (more for wide rockets). Push them down from the top until they sit on the ring at the top of the motor tube. Don't pack them hard.
2. Slide in the rolled parachute. Fold the cord in a zig-zag on top.
3. Push the nose on. Don't trap the cord or lines in the joint.

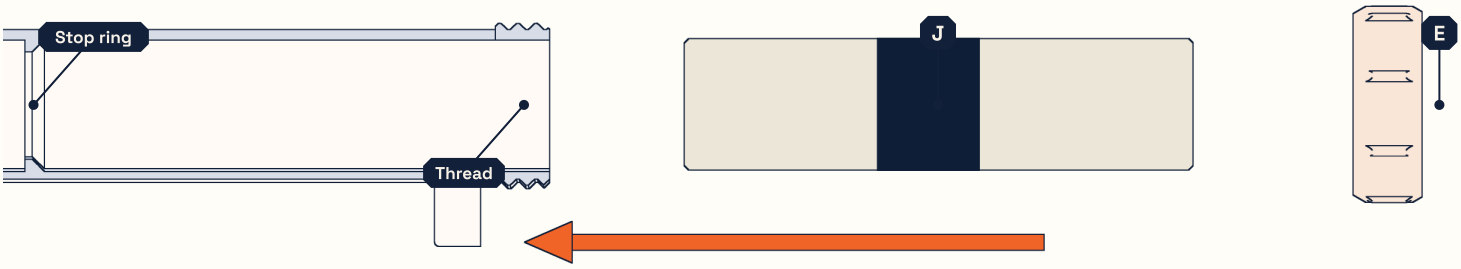
CHECK THE NOSE FIT

Hold the rocket nose-down and shake gently: the nose should stay on. A gentle pull should take it off. Too loose? One wrap of masking tape on the nose's shoulder. Too tight? Sand the shoulder.

ADULT STEPS 9–12

A grown-up handles the motor and igniter. Use only certified model rocket motors, as they come. Never cut, drill, heat or change a motor. Keep motors and igniters away from heat and sparks until launch time.

9 Slide the motor into the tail ADULT STEP

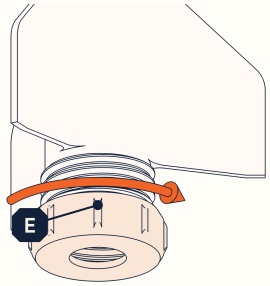


- Check the size: an 18 mm motor for an M18 rocket, and so on.
- Slide it in **nozzle end last**, so the nozzle points out of the tail.
- Push it all the way, until it stops against the ring at the top of the motor tube. That ring takes the motor's push.

TOO TIGHT?

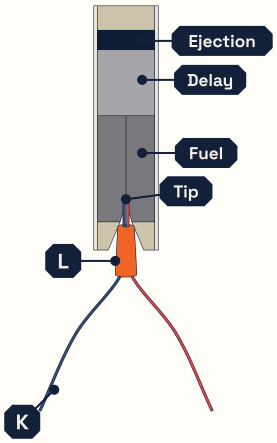
Never force a motor. Look for print strings inside the tube and sand them off. A full-length motor sticks out past the tail by about the height of the thread.

10 Screw on the motor cap ADULT STEP



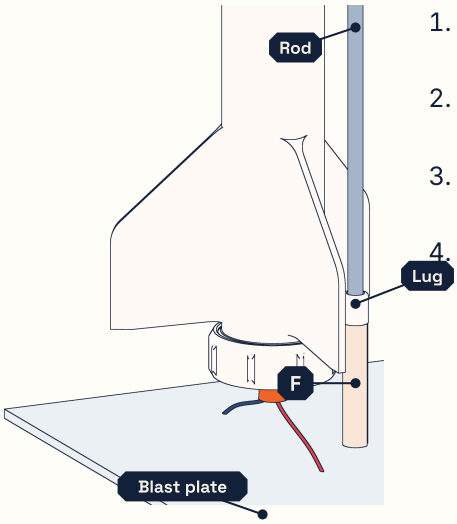
- Turn the cap onto the thread, clockwise as you look at the tail. Start it straight so it doesn't cross-thread.
- **Finger tight** is enough. The cap's floor rests on the end of the motor, so the motor can't be pushed out of the back when the ejection charge fires.
- Never glue or tape the cap on.

11 Put in the igniter ADULT STEP



1. Do this at the pad, with the safety key out of the controller.
2. Push the igniter (K) through the cap's hole into the nozzle, **all the way**, until it touches the fuel.
3. Hold it there and push in the plug (L).
4. Bend the two leads apart so they can't touch.

12 Set up on the pad



1. Snap the rod standoff (F) onto the rod, opening away from the rocket. Slide it down to the blast plate.
2. Slide the rocket down the rod by its lugs until the bottom lug rests on the standoff. The tail is now clear of the plate, with room for the clips.
3. Clip one controller clip to each igniter lead. Clips must not touch each other, the rod or the plate.
4. The rod points straight up (within 30°). Everyone steps back. Key in. Count down: **5, 4, 3, 2, 1, launch!**

PREFLIGHT CHECKLIST — EVERY FLIGHT

- [ ] No cracks. Fins solid. Body joints glued all the way round.
- [ ] Cord tied tight at the strap and the nose. Parachute tied to the loop.
- [ ] New wadding, then the rolled parachute, then the cord. Nose snug, but it comes off.
- [ ] Right motor, pushed all the way in. Cap on, finger tight.
- [ ] **Balance test:** with the motor in, the rocket balances on your finger at the balance point from your report, or nearer the nose. Mark that point with pencil. If it balances behind the mark (nearer the tail), don't fly.
- [ ] The field is big enough (page 7). Wind under 20 mph. No low clouds or aircraft.
- [ ] Launch rod 1/8 in (3.2 mm) unless you changed it in Advanced. Blast plate on. Standoff on the rod.
- [ ] Dry grass and anything that burns cleared from around the pad.
- [ ] The adult in charge keeps the safety key. Everyone at least 15 ft back (30 ft for motors bigger than D).
- [ ] Everyone watching. Count down out loud.

IF IT DOESN'T LAUNCH (MISFIRE)
Take out the safety key. Wait 60 seconds before anyone walks to the rocket. Then check the igniter, or put in a new one.

Fly safely

SAFETY CODE, IN SHORT — NAR

- **Grown-ups handle motors and igniters.** Only certified, factory-made motors, used as the maker says.
- **Electric launch only:** a launch controller with a safety key and a switch that springs back to off.
- **Misfire?** Key out, then wait 60 seconds before anyone goes to the rocket.
- **Launch pad:** a rod or rail with a blast plate, within 30° of straight up. Cap the rod's tip when you're not launching.
- **Size limits:** at most 1500 g at liftoff and 320 N·s of total motor impulse.
- **Where:** outdoors, in an open field, away from buildings, power lines, roads and dry brush. Never launch at targets, into clouds, near aircraft, or with anything that burns as a payload.
- **Recovery:** always a parachute or streamer, with flame-resistant wadding. Never climb trees or power poles to get a rocket back.

This is a summary. Read the full code at nar.org/ModelRocketSafetyCode.

LAUNCH FIELD SIZE — NAR MINIMUM, EACH SIDE

MOTOR CLASS	FEET	METRES
1/4A, 1/2A	50	15
A	100	30
B	200	60
C	400	120
D	500	150
E, F	1000	300

On a breezy day a rocket can drift further than this. Your Rocket Builder report gives the size your rocket needs (“How big a launch field?”). Put the pad near the upwind edge.

AFTER THE FLIGHT

- Let the motor cool. Unscrew the cap. Push the used motor out from the front with a dowel.
- Shake out the burnt wadding. Wipe soot off the cap and thread.
- Check the cord, knots, parachute and joints. Reprint a cracked cap or part.

TROUBLESHOOTING

PROBLEM	TRY THIS
A shoulder or the nose is too tight	Sand off the bulge at its bottom edge (it printed on the bed), then sand lightly all round. Test often.
The nose falls off too easily	One wrap of masking tape round the nose's shoulder. Add more only if needed.
The cord won't go behind the strap	Use thinner cord (up to about 2 mm). Lead it with a bent wire or a floss threader. Clear print strings out of the slot.
The motor won't go in	Is it the right size? Clear strings from the motor tube. Never force it.
The cap won't screw on	Is the motor all the way in? Clean the first turn of the thread. Start the cap straight.
The motor didn't light	Key out, wait 60 s. Check the igniter tip touches the fuel and the clips are clean. Try a new igniter.
The parachute didn't open, or melted	More wadding. Roll the parachute looser. Check the motor's delay (the last number) matches your report.
The rocket wobbled or looped	Do the balance test. Check the score and motor in Rocket Builder. Don't fly again until it's fixed.
The body came apart at a joint	That joint wasn't glued all the way round. Clean it and glue it again.

FLIGHT LOG

DATE	MOTOR	WEATHER	HOW IT FLEW / NOTES